



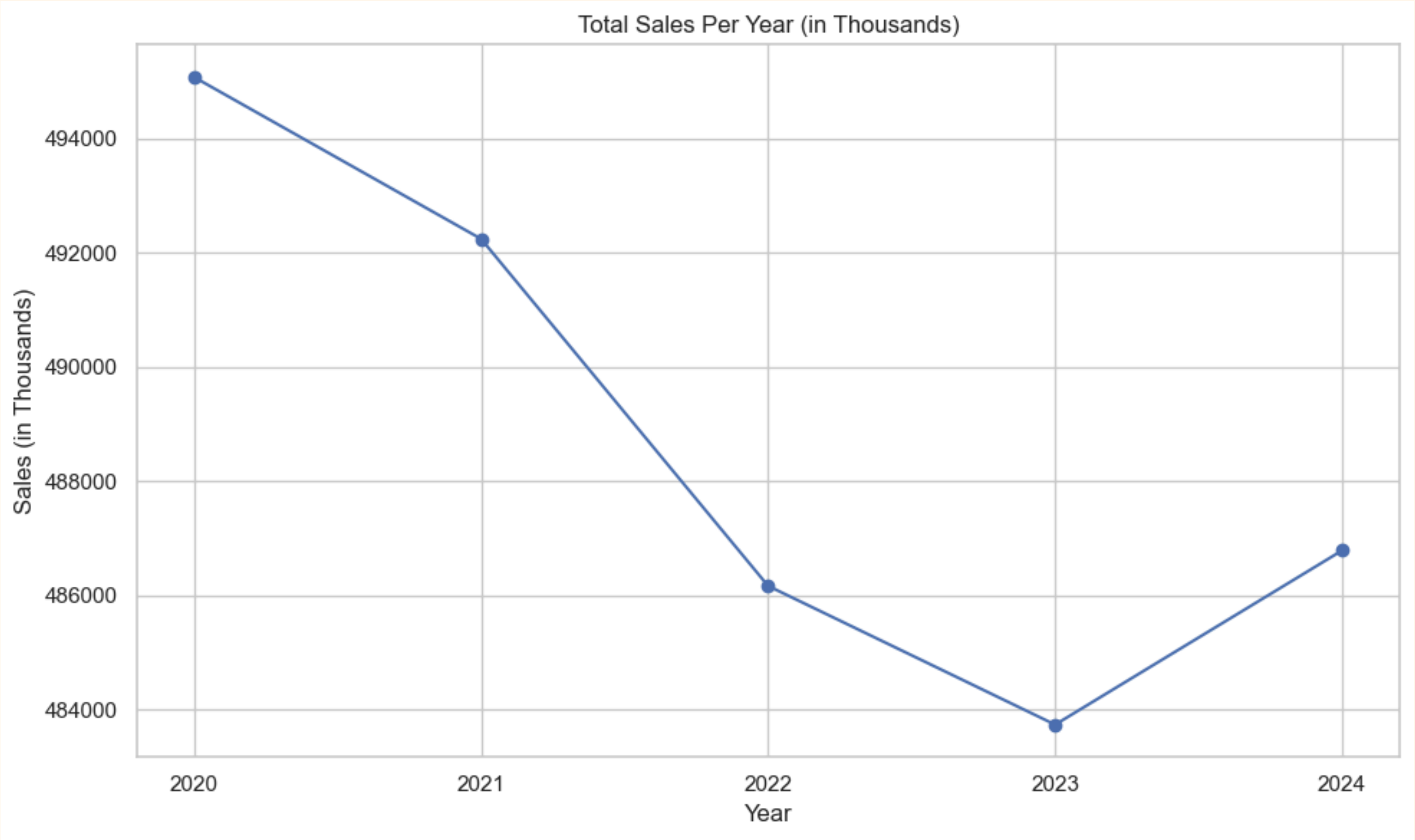
BUSINESS PERFORMANCE REPORT



OBJECTIVE

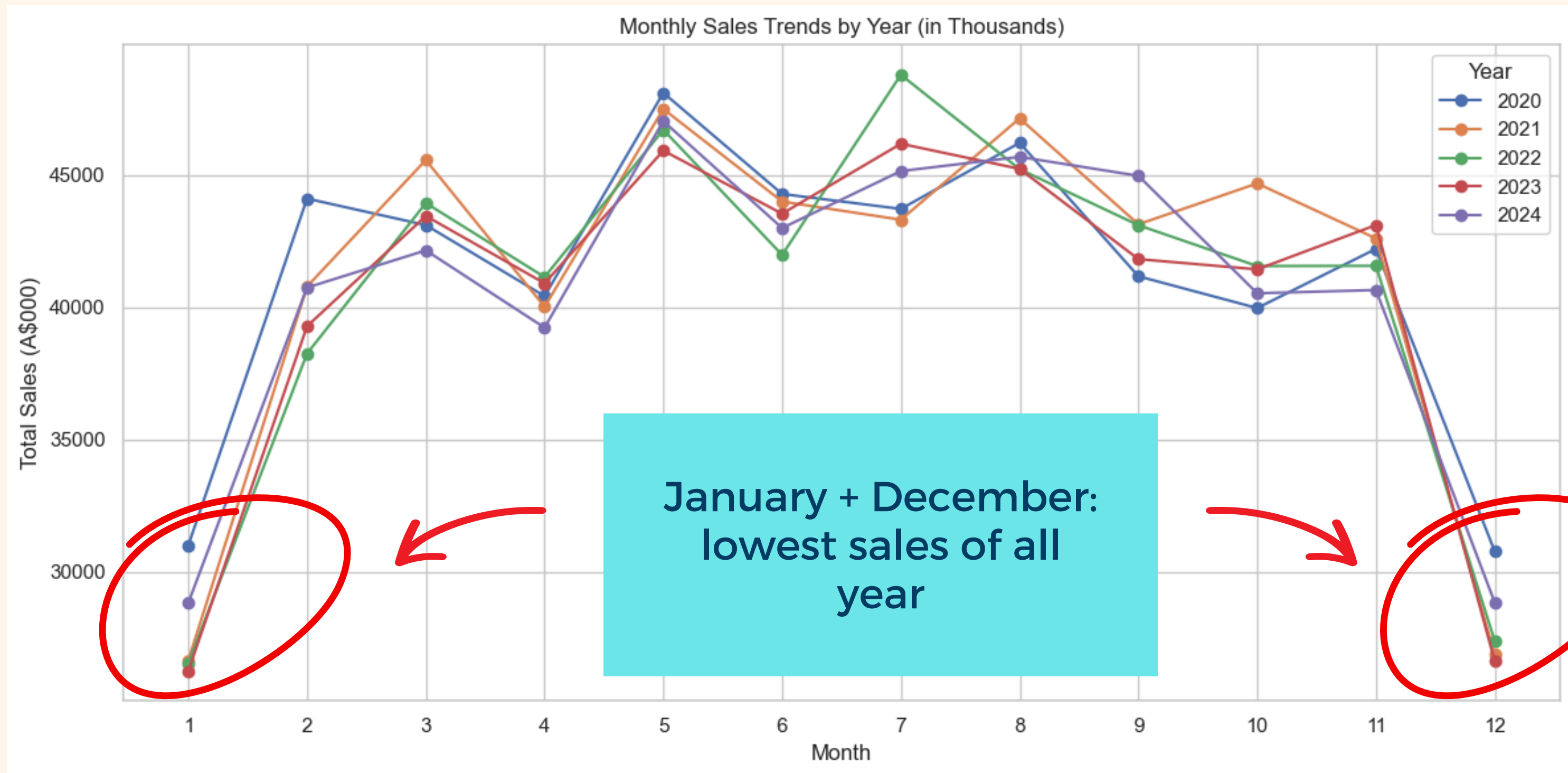
- Identify the key drivers of sales and cost across **customer types, regions, and product lines**
- **Develop data-backed recommendations** to improve growth and operational efficiency

SETTING THE SCENES

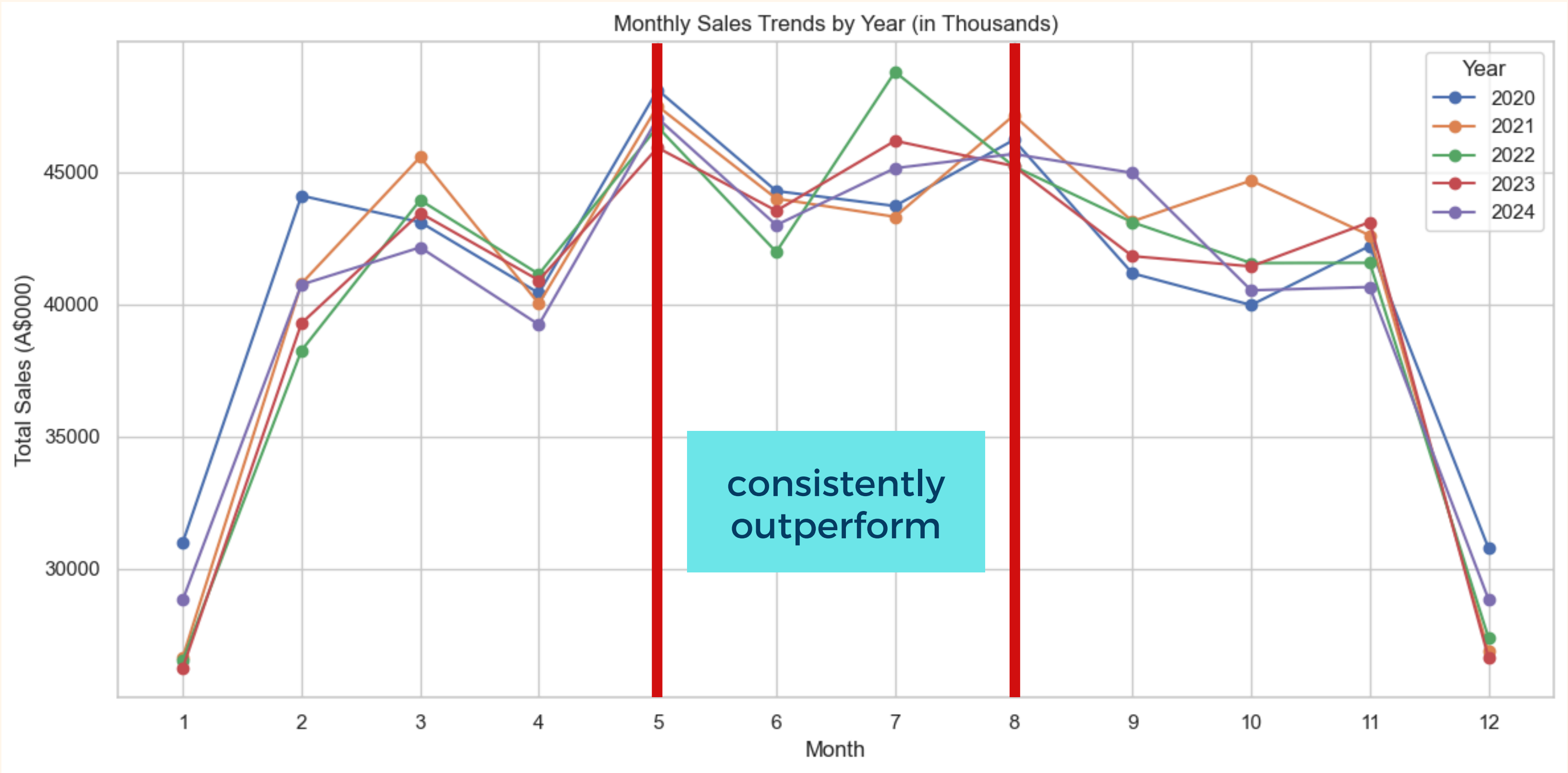


- Annual sales:
- **2020-2023:**
--> Declined continuously
 - **2023-2024:**
--> Minor recovery

A ZOOM IN SALES BY MONTH PER YEAR



A ZOOM IN SALES BY MONTH PER YEAR



GROWTH STRATEGY

Peak Season (May-Aug) - Revenue & Margin Optimization

WHAT WE DO

- Concentrate marketing spend and campaigns in high-demand months
- Limit discounting, prioritise full-price and high-margin products
- Align inventory and supply to peak demand

WHY

- Customers already show strong purchase intent
- Marketing efficiency is highest in this period

HOW IT AFFECTS

- **Higher conversion rates:** more customers buy per campaign dollar ➡ **Maximises marketing ROI** (more revenue per \$ spent)
- **Lower discount dependency:** maintain pricing discipline ➡ **Improves gross margin** (less price erosion)
- **Faster stock movement:** inventory turns quickly ➡ **Strengthens cash flow** (faster inventory-to-cash cycle) and **reduces future clearance risk**

GROWTH STRATEGY

Off-Peak Season (Jan–Feb, Nov–Dec) - Stability & Cost Efficiency

WHAT WE DO

- Re-activate existing customers
- Prioritise high-margin products as profit drivers
- Use low-margin products selectively for bundling and add-ons, traffic generation, inventory clearance
- Design bundles and offers that protect order-level margin

WHY

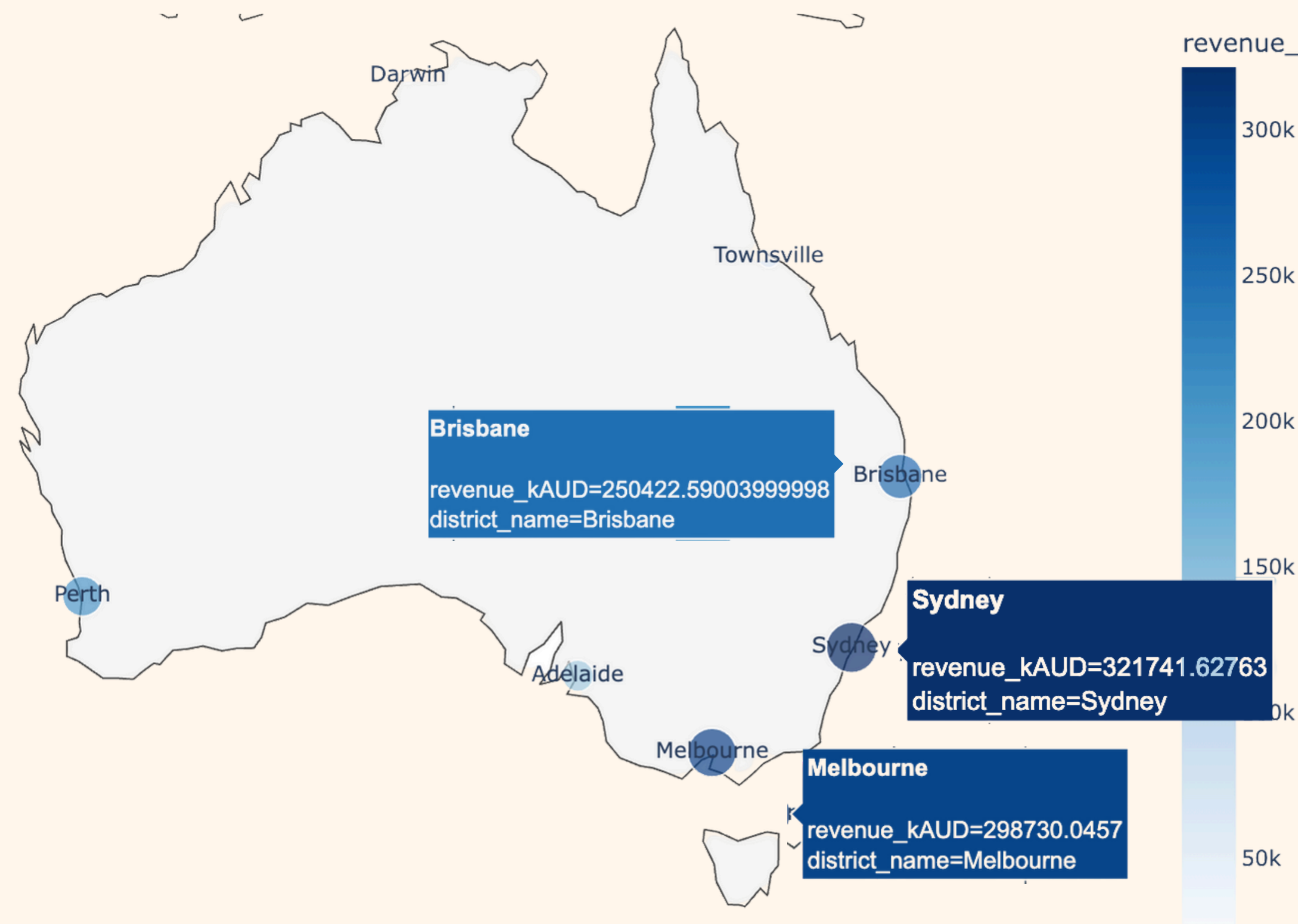
- Incremental off-peak revenue
- Better margin through product mix and cost leverage (High-margin products drive profit; low-margin items drive traffic and support business)
- Lower inventory write-down risk

HOW IT AFFECTS

- **Demand activation:** targeted re-engagement triggers non-planned purchases → **Increase revenue**
- **Price-mix optimisation:** shift volume to higher-contribution products
- **Basket expansion** increases order value → **Faster inventory-to-cash cycle**
- **Inventory acceleration:** move slow stock before deep discounting

GEOGRAPHY MATTERS

WHERE WE WIN, AND WHERE WE COULD WIN MORE



“
BIG CITY = BIG OPPORTUNITY
”

GEOGRAPHY MATTERS

WHERE WE WIN, AND WHERE WE COULD WIN MORE

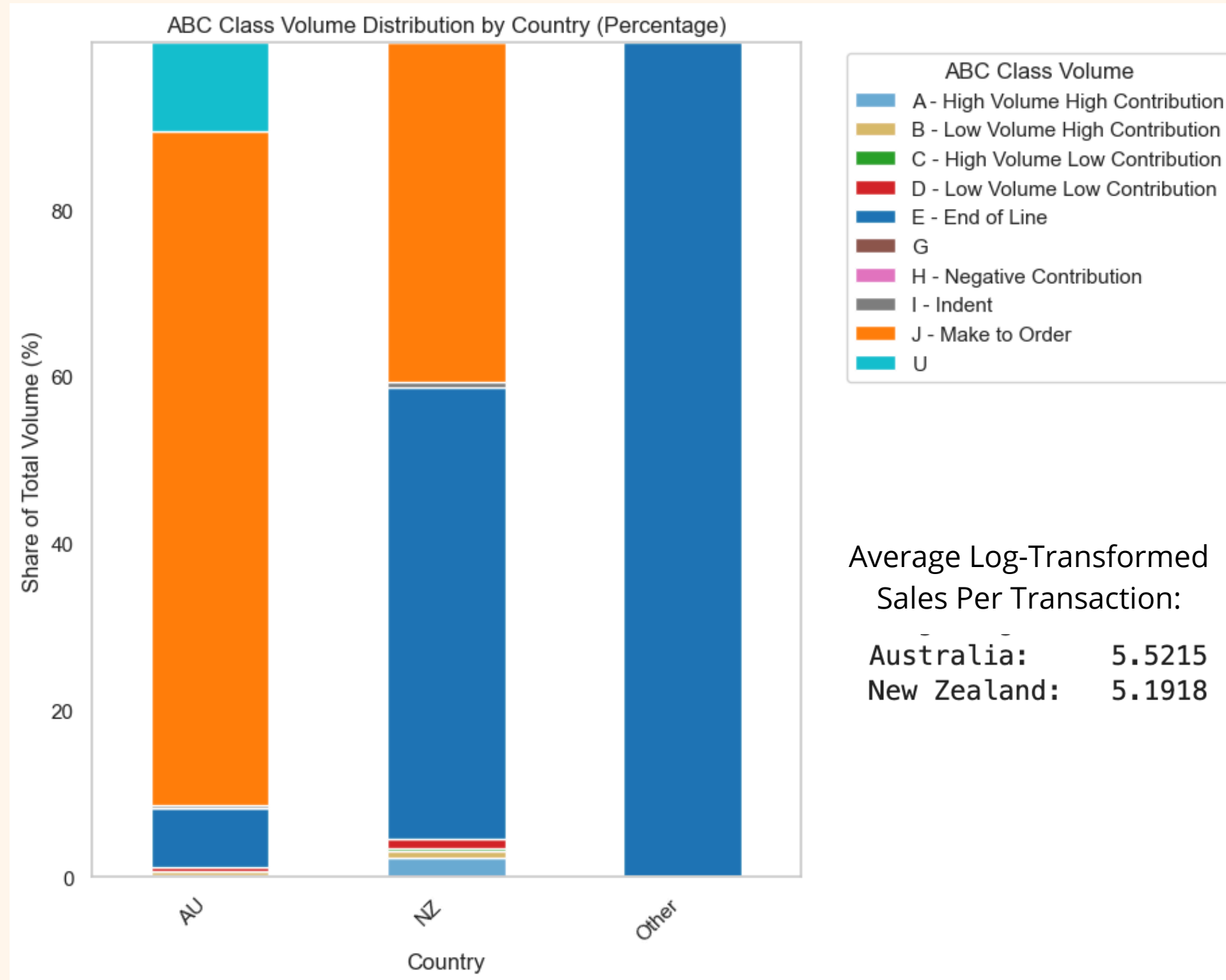
SUMMARY OF SALES PER TRANSACTION TEST	
GROUP	AVG. LOG
SYDNEY + MELBOURNE	5.42
OTHER CITIES	5.61

Sydney & Melbourne combined have lower average sales per transaction

→ Sydney/Melbourne have **higher** transaction **frequency** but **smaller** order **sizes**

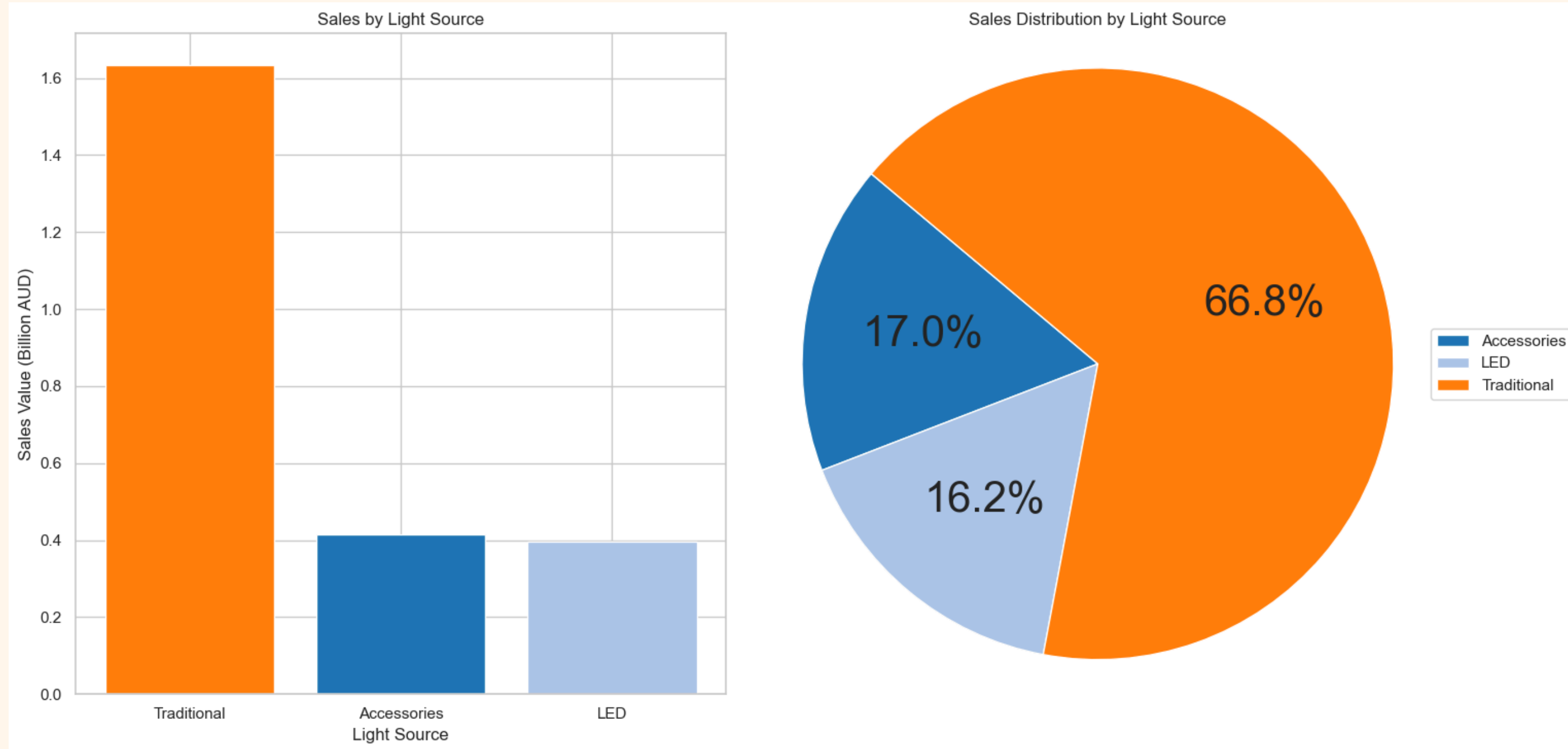
→ Boost basket size through bundling, upselling, or product mix strategies.

WHAT WE SELL SHAPES EVERYTHING



- Australia focuses on A class (high volume, high contribution) and J class (make to order) products
→ Support its higher transaction value
- NZ has a higher share of E-class (end of line - discounted items) and fewer high contribution class,
→ Lower average transaction value
→ Focus on reducing and promoting high-contribution items to improve NZ performance in sales per transaction.

WHAT WE SELL SHAPES EVERYTHING



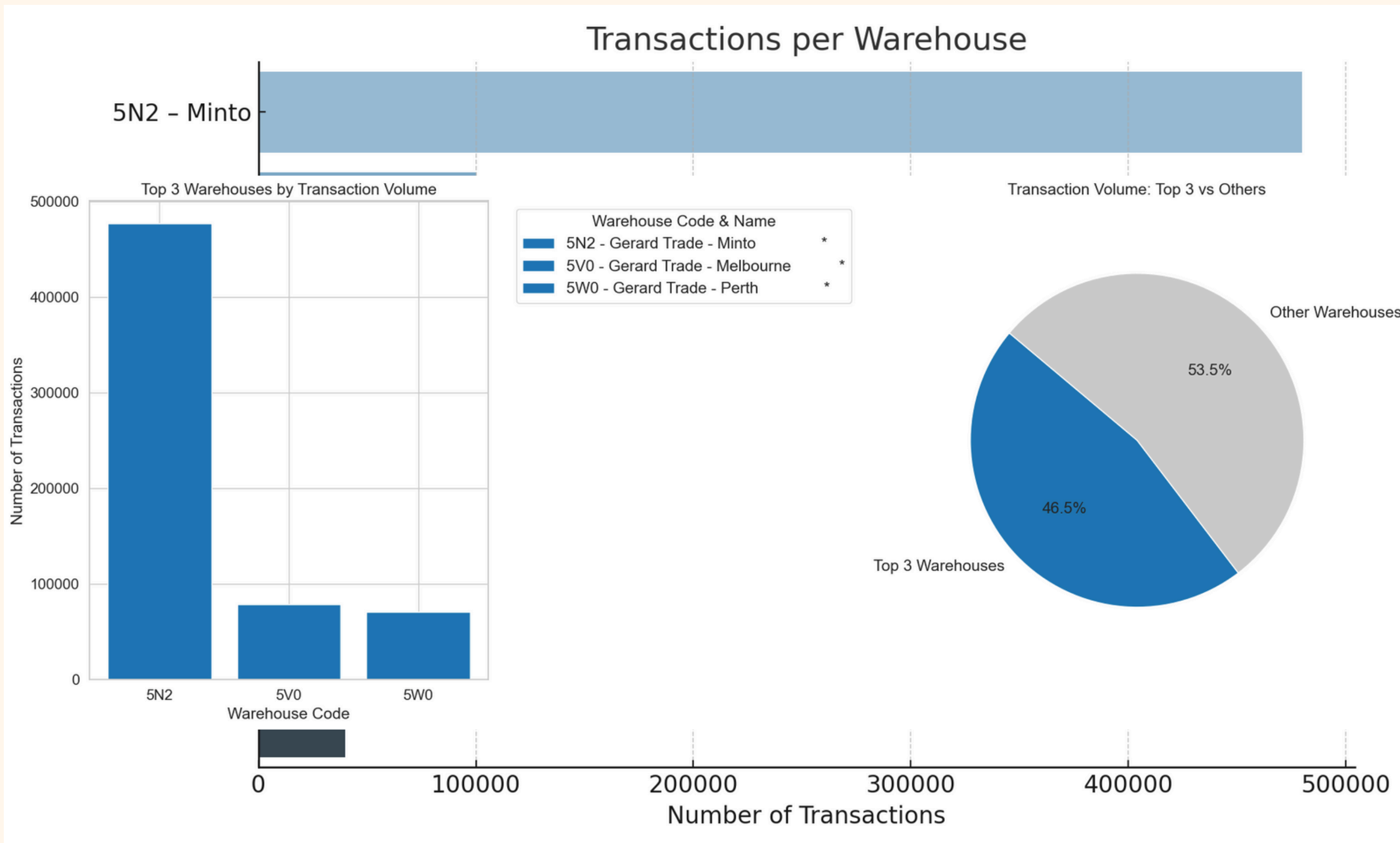
- **HIGHEST PORTION:** Traditional lighting (67% of total sales value)
- **LED and Accessories:** around 16%
→ Opportunities beyond Traditional.

Optimizing product mix, including LED adoption and high contribution class could help improve situation in NZ and boost sales in AUS.

HYPOTHESIS TESTING & OPERATIONAL INSIGHTS:

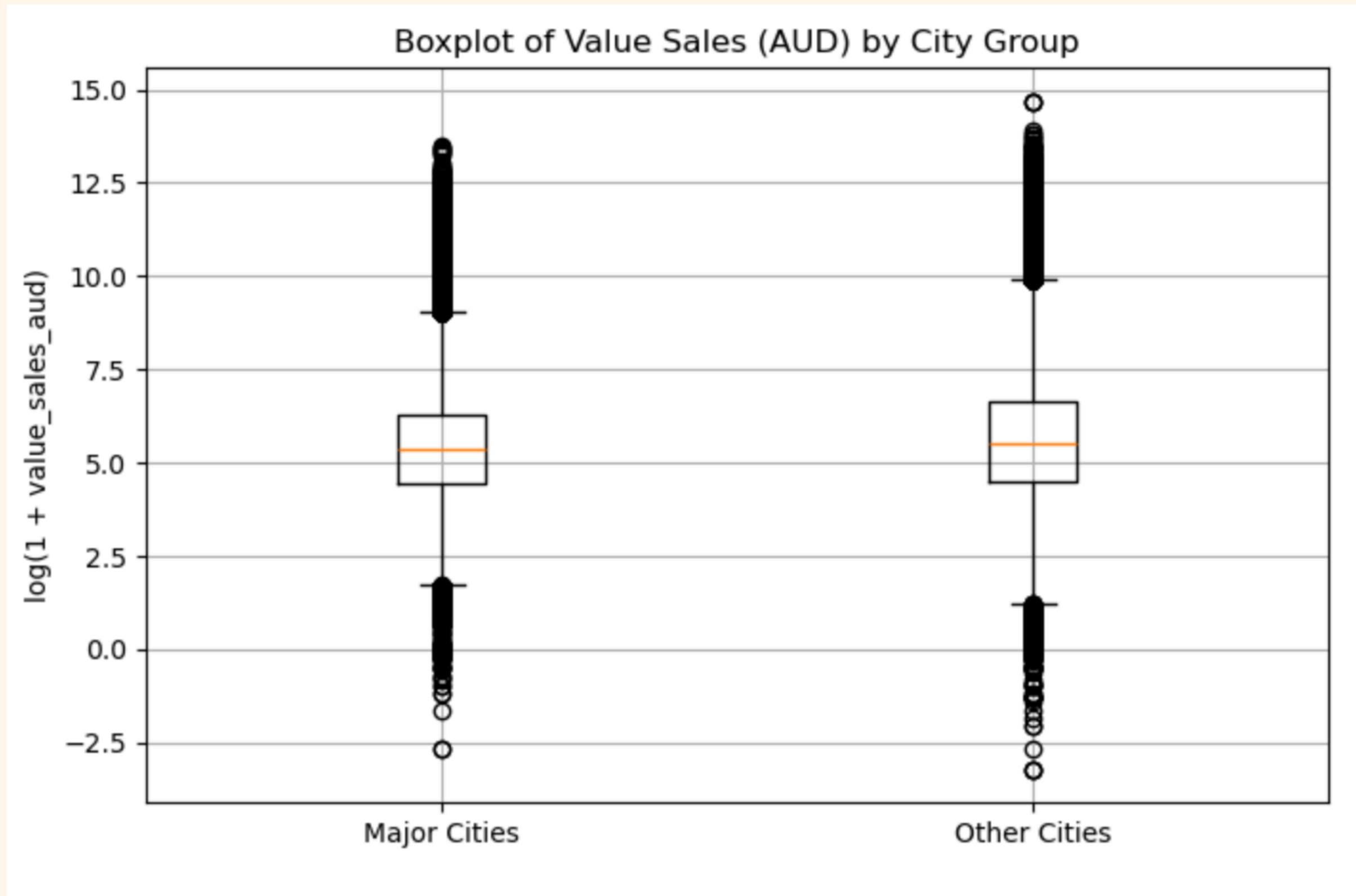
**Validating performance differences across
warehouses, cities, customer and bonus groups**

Warehouse Distribution & Operational Load



- **Warehouse 5N2 – Minto** handles nearly half a million transactions
- **Top 3** warehouses only account for **46.5%** of total volume
- Remaining volume is **fragmented** across **smaller locations**

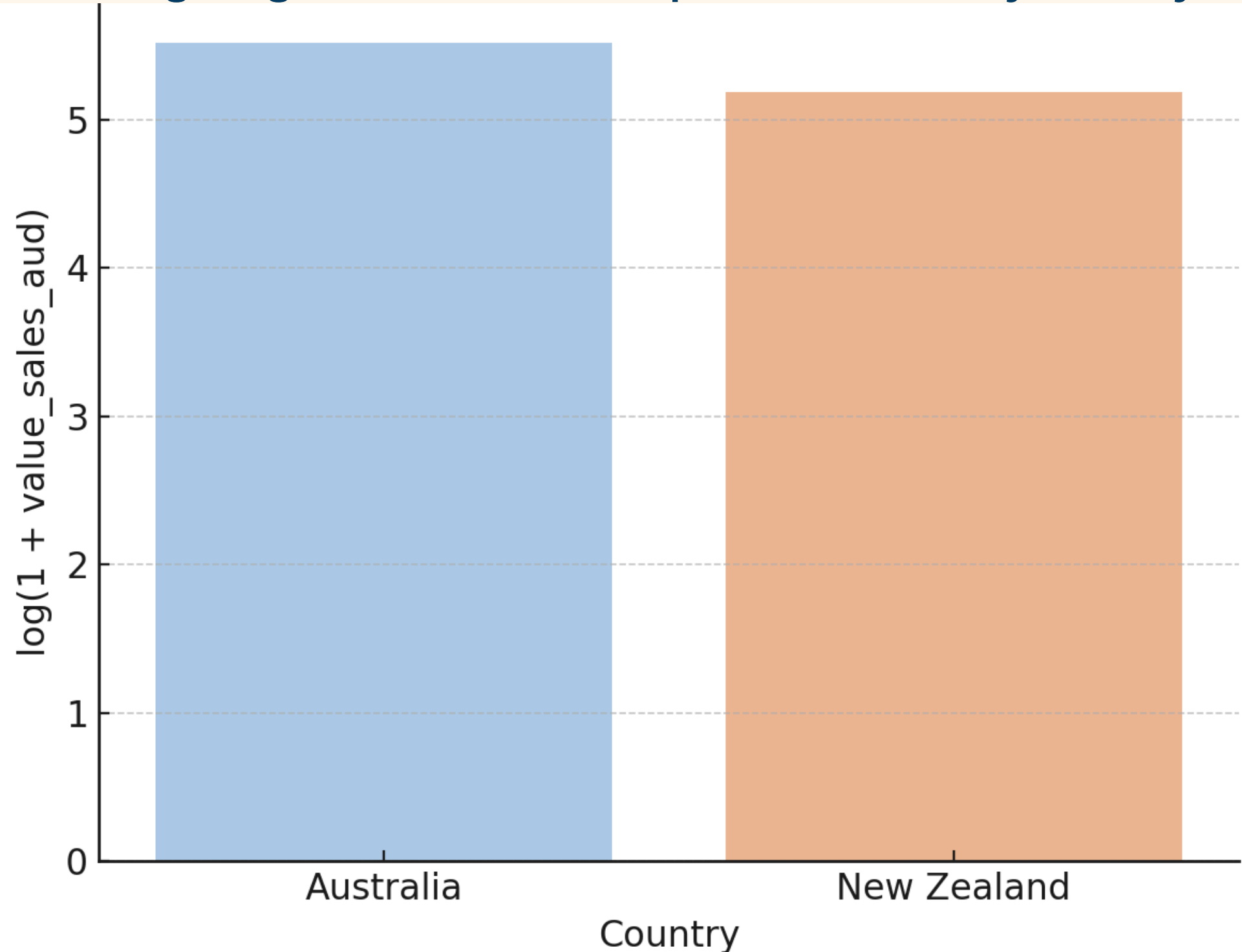
Urban Centers vs. Other Cities



- Two-sample t-test
→ p-value of 0.000
- There is a significant difference in average sales for urban centres versus other cities
- Other cities have higher average sales per transaction

Do Australian Orders Differ from New Zealand?

Average Log-Transformed Sales per Transaction by Country



Average Log-Transformed Sales Per Transaction:

Australia: 5.5215

New Zealand: 5.1918

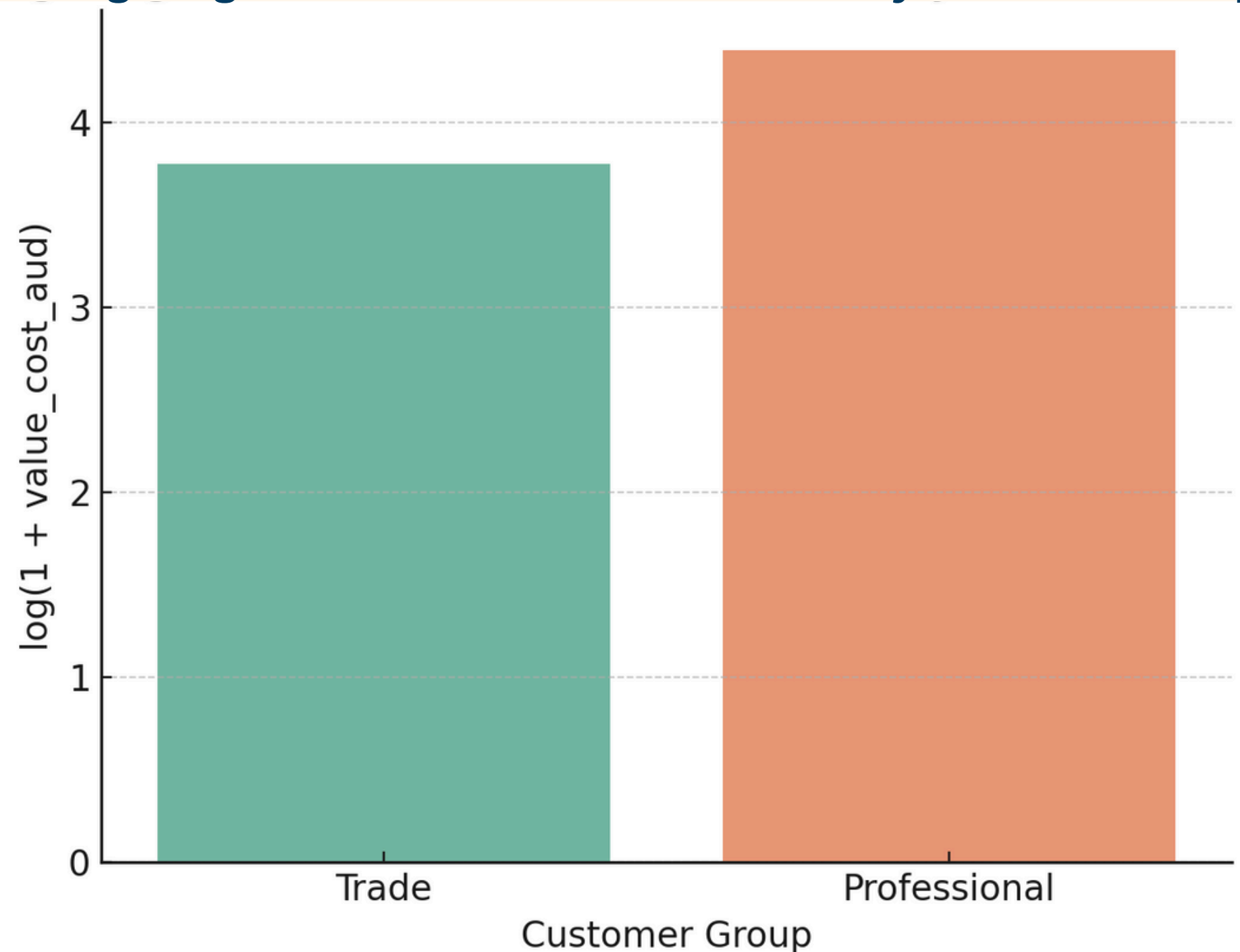
T-statistic: 64.5474489675805

P-value: 0.0

- Using the two-sample t-test we got a p-value of 0.000
- Australian orders or transactions are larger in value
- There is a significant difference between orders from Australia and New Zealand

Trade vs. Professional: Cost Comparison

Average Log-Transformed Transaction Cost by Customer Group



Average Log-Transformed Transaction Cost:
Trade: 3.7737
Professional: 4.3885

T-statistic: -141.84671011027706
P-value: 0.0

- Two-sample t-test
--> p-value = 0.000
- Professional bonus groups are more costly

Cost + Volume = Fulfillment Pressure

- Reinforces the idea: low-value, high-volume orders = operational bottleneck
- >> Warehouse design and customer segmentation strategies should be aligned.

MULTIPLE REGRESSION

Uncover What Drives Performance

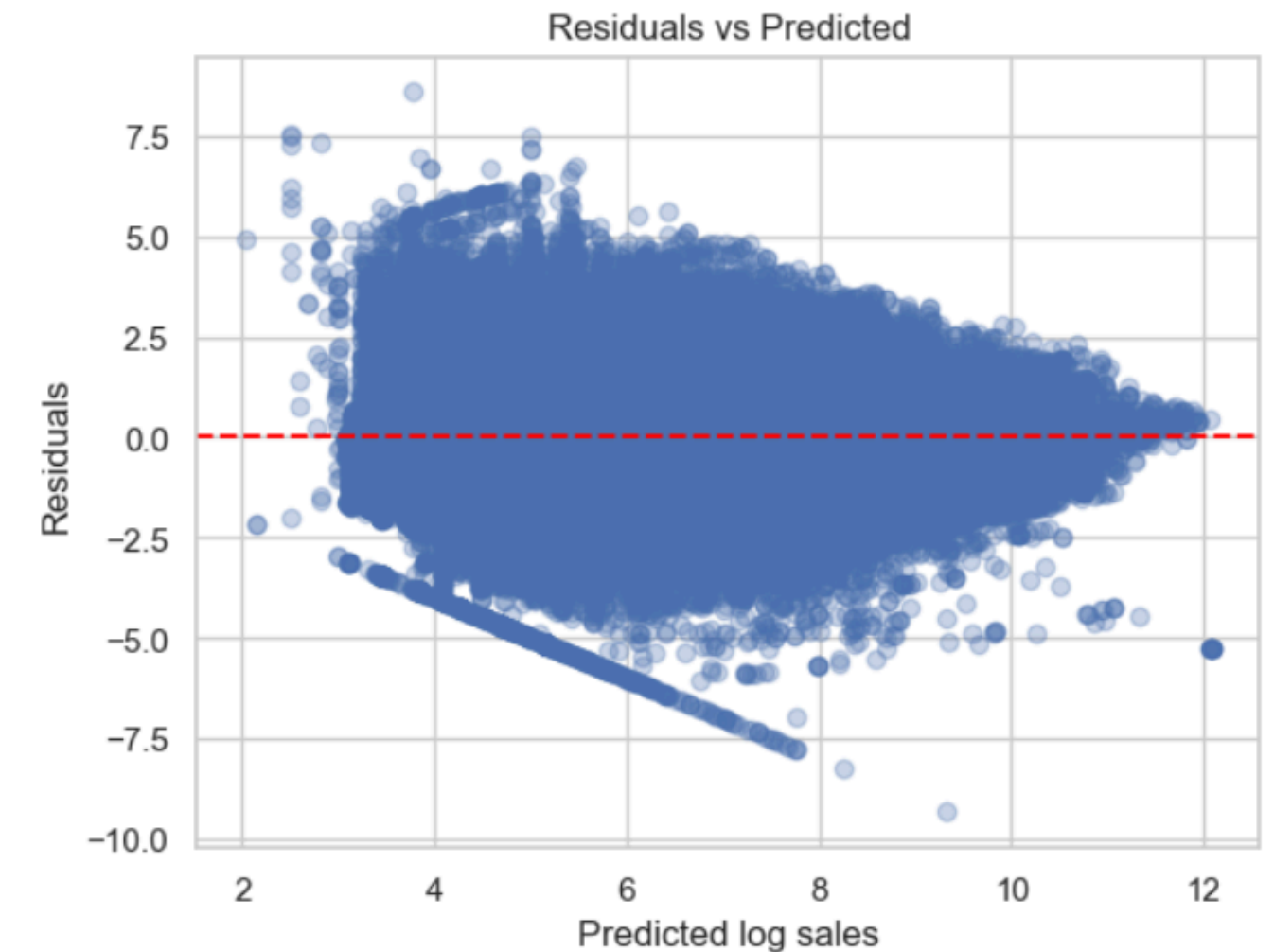
LET THE MODELS SPEAK WHAT DRIVES SALES?

OLS Regression Results

```
=====
Dep. Variable:          y      R-squared:          0.601
Model:                  OLS    Adj. R-squared:      0.601
Method:                 Least Squares    F-statistic:    6.670e+04
Date:                  Thu, 29 May 2025    Prob (F-statistic):    0.00
Time:                  21:57:40    Log-Likelihood:    -2.3845e+06
No. Observations:      1769701    AIC:              4.769e+06
Df Residuals:          1769660    BIC:              4.770e+06
Df Model:               40
Covariance Type:       nonrobust
=====
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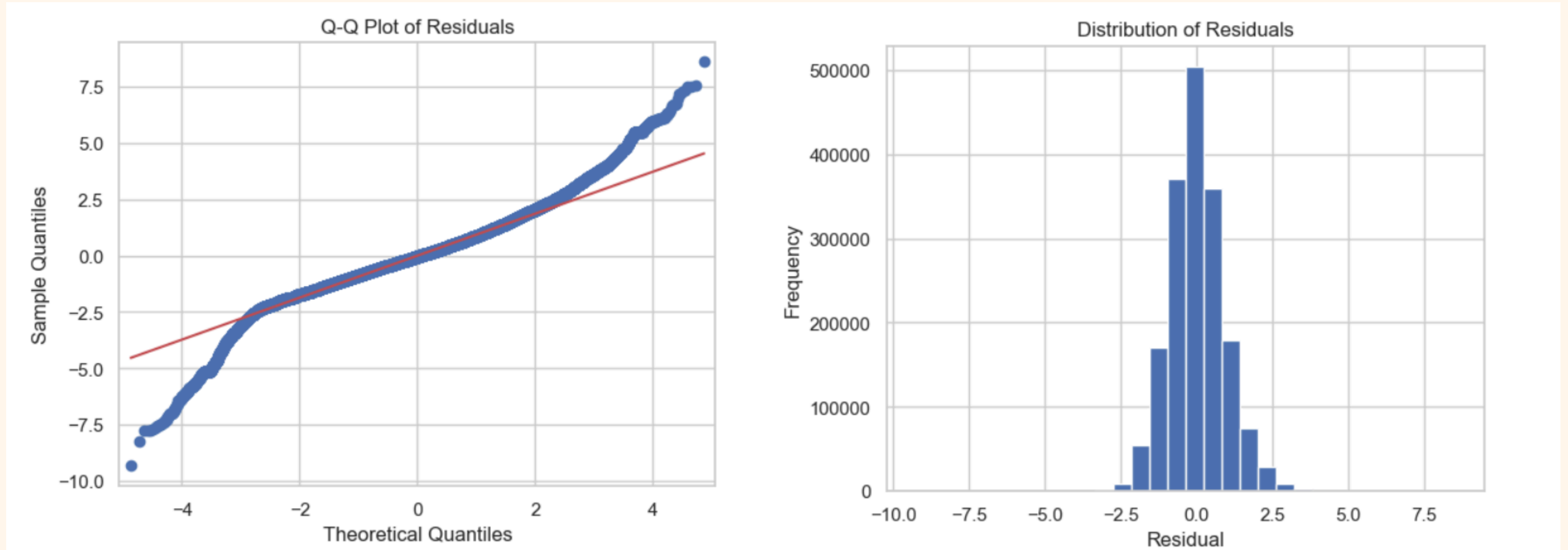
Sales Model

- **Adjusted $R^2 = 0.601$ → strong explanatory power**
- Top drivers: Item type, business area, month, bonus group
- Confirms: Customer type and item mix matter more than seasonality or region



- Residuals are mostly well spread
- Possible linear pattern at lower end → model less accurate for small sales
- Suggests heteroscedasticity or hidden structure → worth further checking

LET THE MODELS SPEAK WHAT DRIVES SALES?



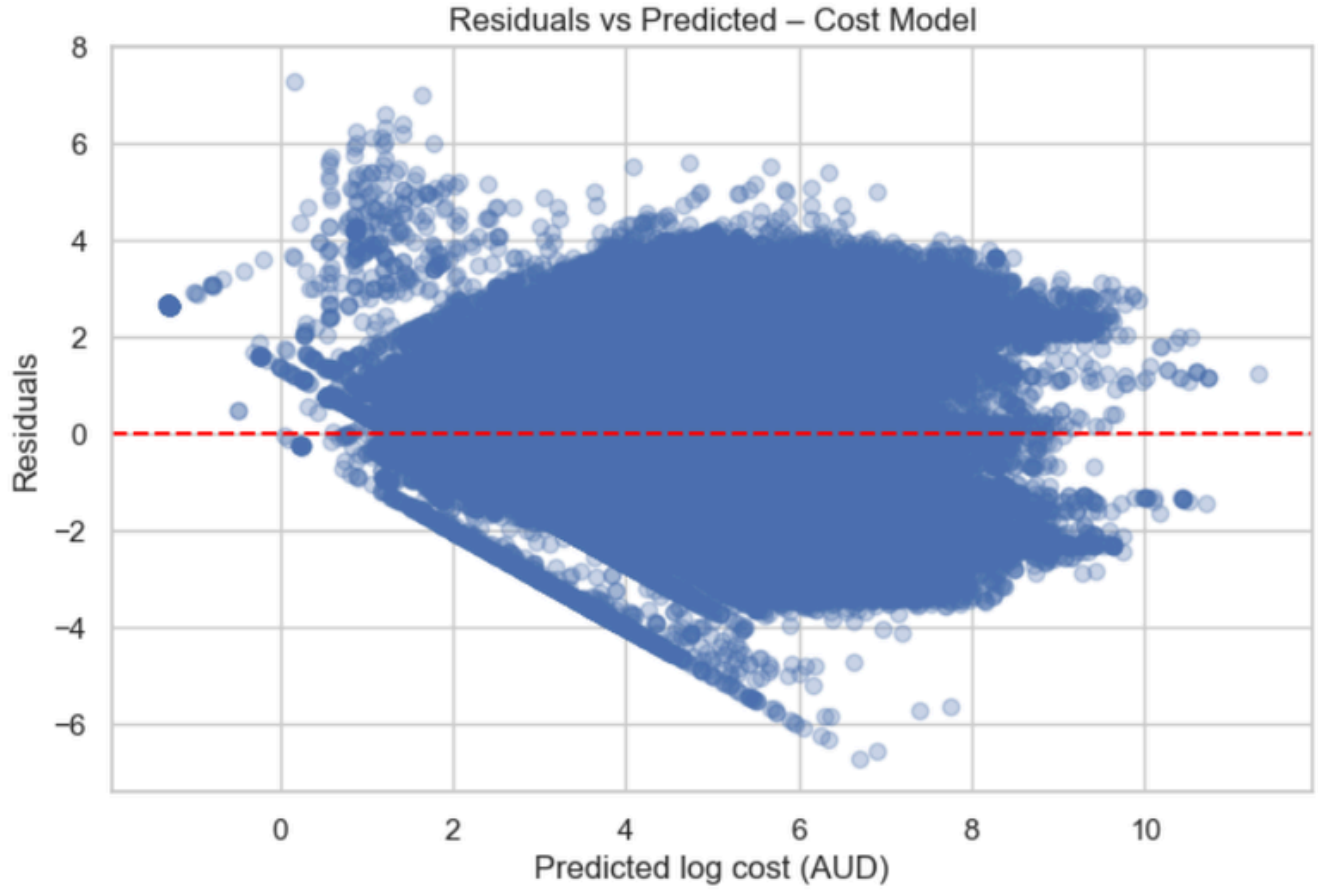
We also checked the normality of residuals, which is a key assumption in linear regression.

- Q-Q plot: most points follow the line → residuals are close to normal.
- Histogram: bell-shaped curve centered at zero.
- Confirms the model meets key assumptions.

LET THE MODELS SPEAK WHAT DRIVES SALES?

OLS Regression Results

Dep. Variable:	y	R-squared:	0.433
Model:	OLS	Adj. R-squared:	0.433
Method:	Least Squares	F-statistic:	3.245e+04
Date:	Thu, 29 May 2025	Prob (F-statistic):	0.00
Time:	22:16:00	Log-Likelihood:	-2.7229e+06
No. Observations:	1702692	AIC:	5.446e+06
Df Residuals:	1702651	BIC:	5.446e+06
Df Model:	40		
Covariance Type:	nonrobust		

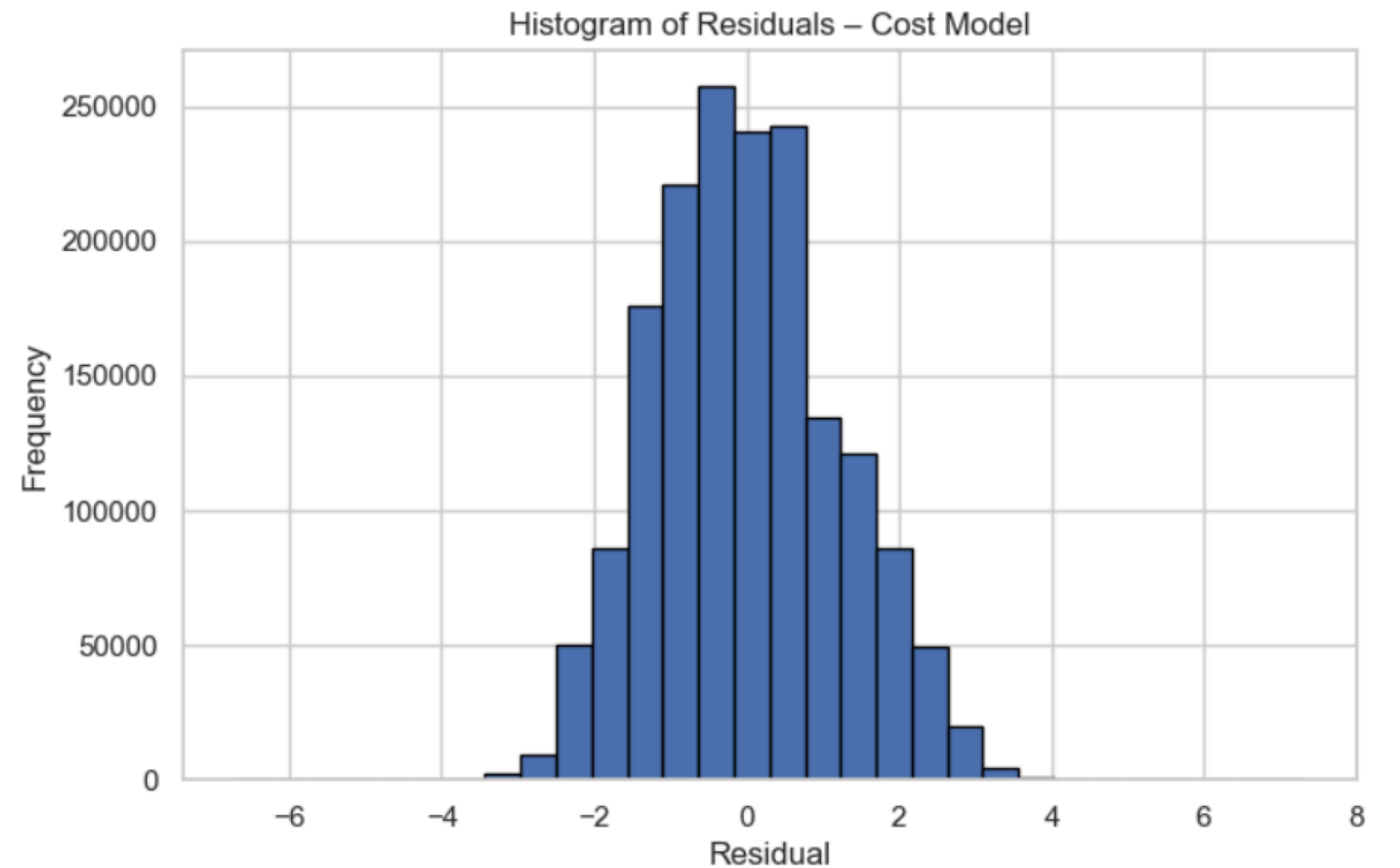
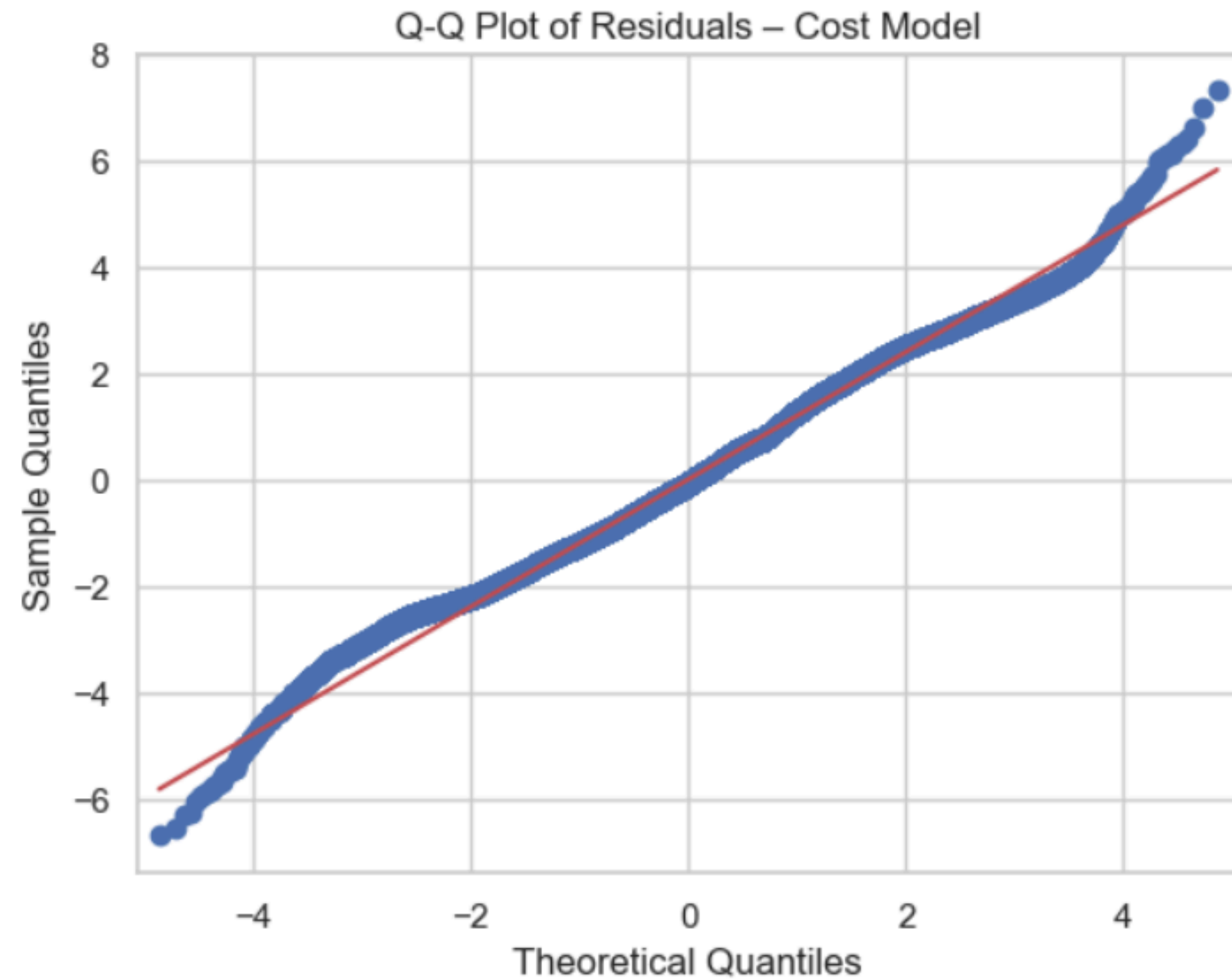


Cost Model

- **Adjusted $R^2 = 0.433 \rightarrow$ moderate fit**
- Top factors: Business area, item type, trade group
- Trade group $\rightarrow$ strongest influence on cost
- Model assumptions hold (normality, homoscedasticity)
 $\rightarrow$ results are reliable

- Residuals spread well, but show a downward linear trend at low predicted cost
- This means the model may underestimate small-cost cases
- Suggests mild non-linearity or outliers $\rightarrow$ worth deeper look

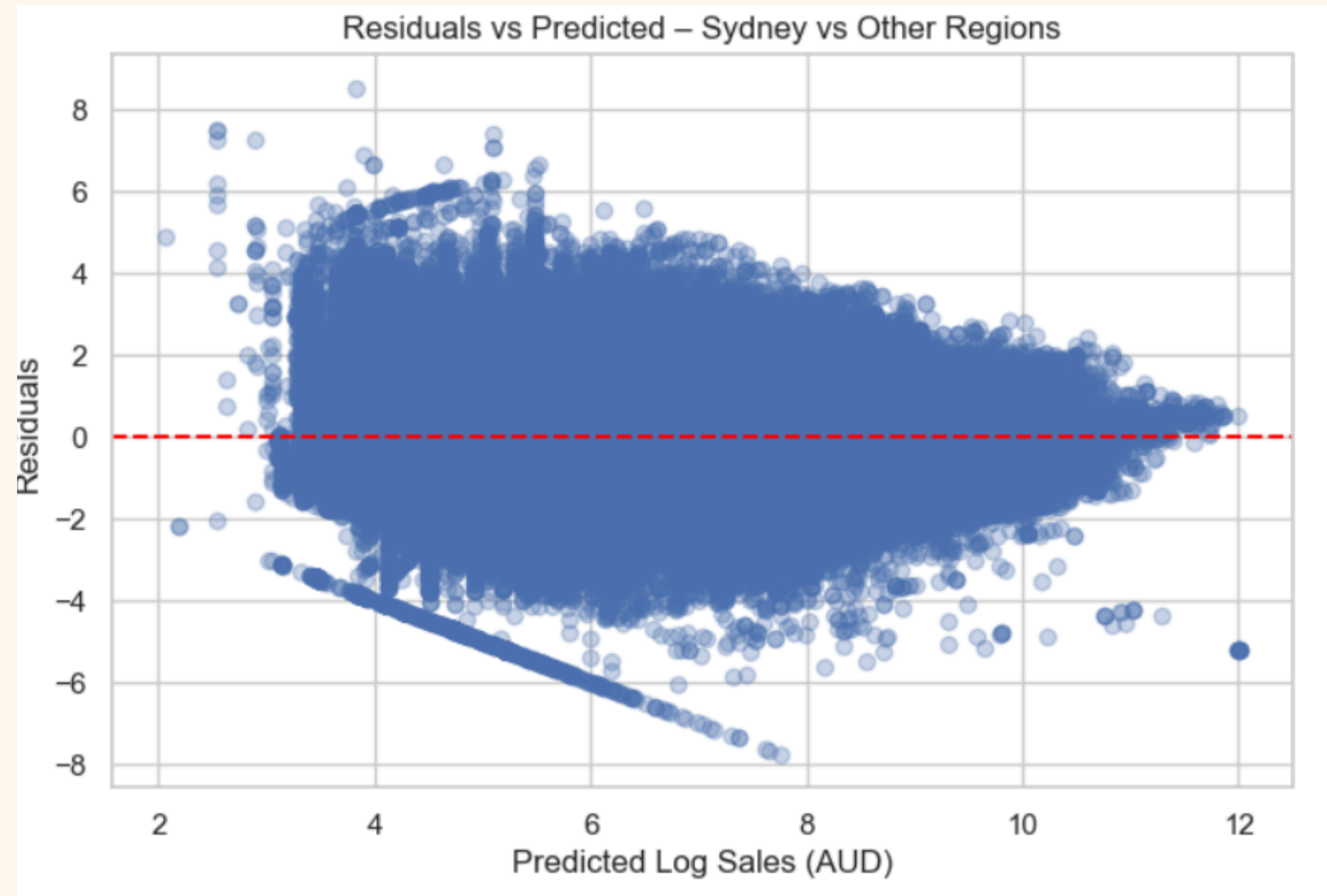
LET THE MODELS SPEAK WHAT DRIVES SALES?



- Q-Q plot: residuals mostly follow a straight line → normal
- Histogram: symmetric distribution around 0
- Confirms model meets linear regression assumptions

LET THE MODELS SPEAK WHAT DRIVES SALES?

WHY IS SYDNEY SO POWERFUL IN TOTAL REVENUE, BUT WEAK IN TRANSACTION VALUE?



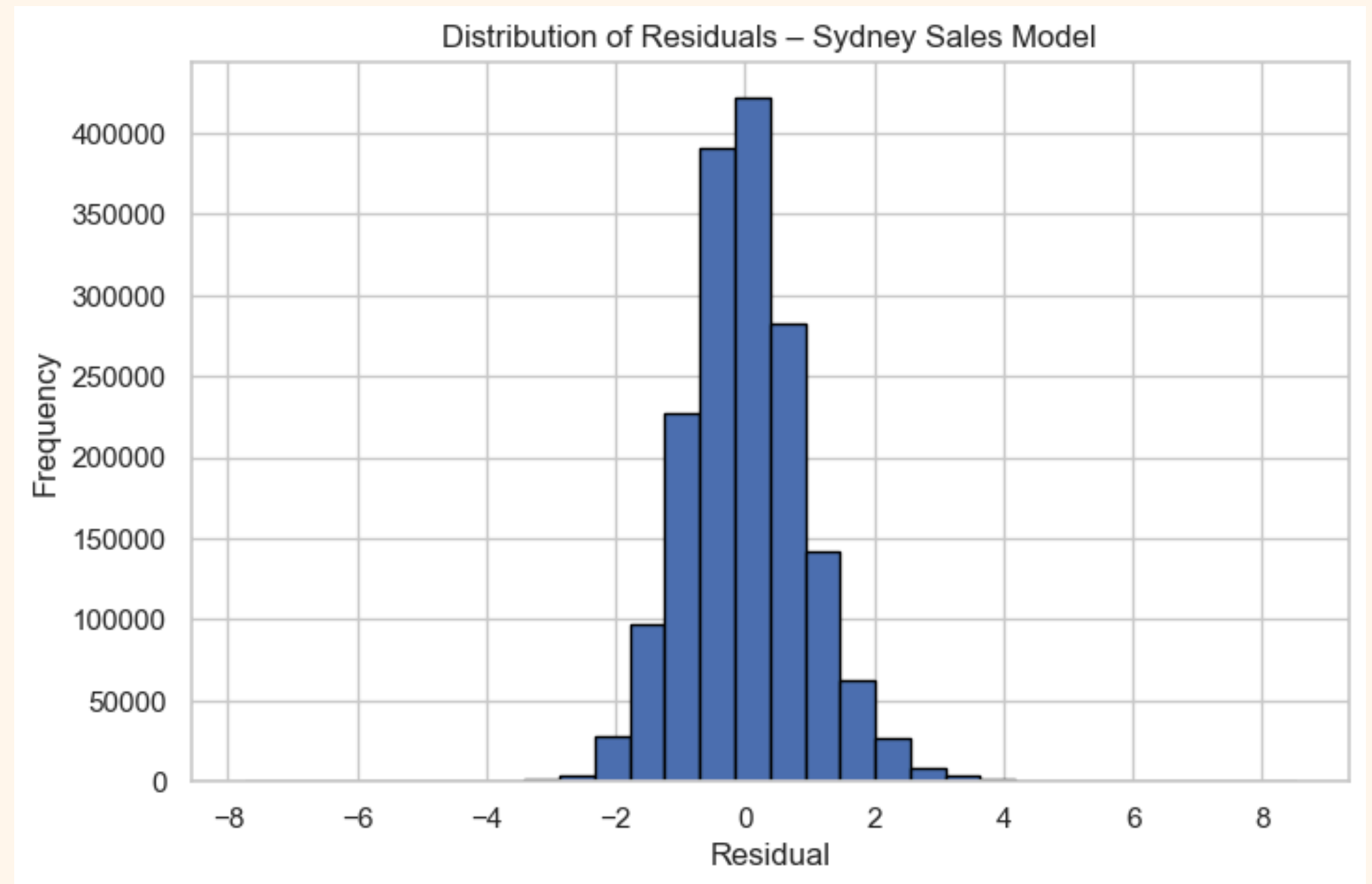
- Model struggles more with small sales values
- The “triangle shape” shows less accuracy at low sales
→ Likely from Sydney's small transactions
- Diagonal gap suggests the model may miss unique traits in Sydney: bundle size, customer type, or purchase style

LET THE MODELS SPEAK WHAT DRIVES SALES?

WHY IS SYDNEY SO POWERFUL IN TOTAL REVENUE, BUT WEAK IN TRANSACTION VALUE?

- The shape is mostly symmetric and centered around zero.
 - The model is not consistently biased Sydney's sales too often.
- The assumption of normality is reasonably met → one of the requirements for trusting our regression results.

--> Most prediction errors are small and balanced, meaning the model is generally accurate and trustworthy.



SUMMARY

- **Sydney has high revenue sales but low average transaction size.**
- **Significant differences in sales between Australia and New Zealand**
- **Professional bonus group has high costs**

RECOMMENDATIONS

- **Launch bundle deals or upselling promotions in Sydney**
 - Leverage Sydney's high revenue sales to boost average transaction size
- **Tailor regional marketing and pricing strategies**
 - Aligning performance differences across countries, and cities to boost average value per sale.
- **Optimize cost management for the Professional bonus group**
 - Assess if operational refinement is needed